

Multiversal entropy and information conservation in black hole nucleated baby universes

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ABSTRACT

We propose a novel mechanism by which supercritical black holes formed via vacuum bubble or domain wall nucleation during the inflationary epoch not only give rise to baby universes, but also contribute to a generalized entropy accounting in the multiverse. We show that the interiors of such black holes inflate eternally and become new domains within the multiverse, connected via non-traversable wormholes. We conjecture that these disconnected regions remain quantum mechanically entangled with the parent universe. This entanglement modifies the Bekenstein-Hawking entropy bound, implying a multiversal generalization of the second law of thermodynamics. Our results suggest that the total entropy of the multiverse increases not only due to local thermodynamic evolution, but also via creation of entangled inflating domains beyond causal contact.

1. Introduction

The interplay between black hole physics and cosmology has long been at the heart of our understanding of the universe's structure, thermodynamics, and ultimate fate. In particular, the synthesis of quantum field theory, general relativity, and inflationary cosmology provides a fertile ground for exploring profound questions related to the origin of the multiverse, the nature of singularities, and the conservation of information. The seminal discovery that black holes emit radiation due to quantum effects near the event horizon [1] initiated a cascade of developments probing the thermodynamic and quantum mechanical behavior of horizons. These insights have been further sharpened by the formulation of the generalized second law [2,3] and the holographic principle [4–6].

In the context of early universe cosmology, the inflationary paradigm [7–9] not only resolves key puzzles such as the horizon and flatness problems, but also predicts a stochastic background of quantum fluctuations that seed cosmic structure [10,11]. Moreover, inflation generically gives rise to eternal inflation [12,13], in which different regions of spacetime inflate at different rates or exit inflation at different times, producing a multiverse of causally disconnected domains. The mechanism for this proliferation is rooted in the quantum tunneling between metastable vacua in the landscape of scalar field potentials—a framework further developed in string theory and the so-called string landscape [14–17].

Recent work by Garriga, Vilenkin, and Zhang [18] has introduced a compelling extension to this picture, demonstrating that vacuum bubbles or spherical domain walls nucleated during inflation can evolve into black holes after inflation ends. Depend-

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ing on their mass, such black holes may either collapse to singularities or develop wormhole structures connecting the ambient Friedmann-Robertson-Walker (FRW) universe to an inflating baby universe. This scenario opens a new avenue for multiverse realization, where each sufficiently large black hole becomes the seed of a causally disconnected domain with its own cosmological history. Similar ideas were previously considered in various forms [19–21], but the detailed connection between black hole mass thresholds and internal inflation has only recently been clarified.

While many foundational works on black hole-induced baby universes date back more than a decade, there has been substantial progress in the last few years that strengthens the motivation for our framework. Parallel developments in quantum gravity and holography emphasize the role of wormhole geometries and entanglement entropy in preserving unitarity across disconnected domains [22,23]. On the observational side, analyses of gravitational-wave echoes and constraints on exotic compact objects have advanced significantly [24], and updated surveys of primordial black hole populations continue to refine the mass windows relevant to inflationary bubble collapse [25]. These works collectively highlight that the questions addressed in this paper remain at the forefront of both theoretical and observational research.

In this paper, we build upon these results to propose a unifying framework wherein black holes formed via bubble nucleation act as conduits for multiversal branching. Crucially, we posit that the baby universes thus formed remain quantum mechanically entangled with their progenitor regions through correlations imprinted at the time of black hole formation. This leads us to a modification of the Bekenstein-Hawking entropy formula, incorporating a multiversal entanglement entropy term. The implications are twofold: first, information conservation in black hole evaporation may be preserved across universes, and second, the second law of thermodynamics must be generalized to account for entropy flux into causally disconnected domains.

Such a viewpoint echoes and extends the ideas proposed in the black hole complementarity [26], firewall debates [27], and ER=EPR conjecture [28], suggesting that quantum entanglement and wormhole geometry are intimately related. Furthermore, our framework provides a novel perspective on the information paradox [29,30], suggesting that rather than information being destroyed, it is transferred nonlocally across the multiverse through entangled baby universes. The implications of this proposal are vast, ranging from refined entropy bounds in cosmology [31,32] to testable signatures in primordial black hole distributions [33–35] and possible quantum gravitational echoes [36].

The rest of this paper is organized as follows: in Section 2, we review the gravitational collapse of vacuum bubbles and the conditions under which inflating baby universes emerge. In Section 3, we develop the concept of multiversal entanglement entropy and derive a modified entropy formula. Section 4 explores the thermodynamic implications, including a proposed multiversal second law. In Section 5, we discuss observational prospects and the broader implications for quantum gravity. Finally, Section 6 summarizes our conclusions and outlines directions for future work.

2. Black hole induced baby universes

The formation of black holes from vacuum bubbles nucleated during inflation provides a natural mechanism for seeding disconnected regions of spacetime that can subsequently undergo their own inflationary evolution. As discussed in [18], if a vacuum bubble with internal vacuum energy density $\rho_b > 0$ nucleates in a metastable inflating background, it begins to expand due to the pressure difference across the bubble wall. The acceleration is driven by the mismatch in vacuum energy densities between the false vacuum exterior (ρ_i) and the lower-energy true vacuum interior ($\rho_b < \rho_i$), with the dynamics governed by Israel's junction conditions [37] and the thin-wall formalism [38,39].

During inflation, the bubble grows rapidly, and if it nucleates sufficiently early, its physical radius at the end of inflation may exceed the Hubble length associated with the internal vacuum, i.e., $R > H_b^{-1}$, where $H_b = \sqrt{8\pi G\rho_b/3}$. After inflation ends, the energy of the inflaton is transferred to a radiation or matter-dominated fluid, depending on the post-inflationary reheating dynamics. The expanding bubble, now embedded in this cosmological background, interacts gravitationally and kinetically with the ambient fluid. The dissipation of the bubble wall's kinetic energy through momentum transfer leads to gravitational collapse [18,38]. The resulting spacetime is no longer homogeneous: it features a localized Schwarzschild geometry nested within a Friedmann-Robertson-Walker (FRW) universe, with the bubble wall acting as the boundary between the two regions.

To formalize the criteria for baby universe formation, we consider the dynamics of a vacuum bubble embedded in an FRW background, transitioning into a Schwarzschild-de Sitter (SdS) geometry after inflation ends. The motion of the bubble wall can be described using the thin-shell formalism and junction conditions. Following [18,39], the equation governing the bubble radius $R(\tau)$ as a function of proper time τ on the wall is,

$$M = \frac{4\pi}{3} R^3 (\rho_b - \rho_0) + 4\pi\sigma R^2 \sqrt{\dot{R}^2 + 1 - H_b^2 R^2} - 8\pi^2 G\sigma^2 R^3, \quad (1)$$

where M is the Schwarzschild mass measured by external observers, ρ_b is the vacuum energy inside the bubble, ρ_0 is the (assumed negligible) energy density outside after inflation, σ is the tension of the bubble wall, $H_b = \sqrt{8\pi G\rho_b/3}$ is the Hubble rate inside the bubble, and $\dot{R} = dR/d\tau$ is the radial velocity of the wall in proper time. This relation acts as an effective energy conservation equation. The fate of the bubble depends on whether the solution to Eq. (1) admits a turning point (i.e., $\dot{R} = 0$), indicating collapse, or allows for unbounded expansion, leading to internal inflation.

Eq. (1) can be derived from Israel's junction conditions applied to a spherically symmetric vacuum bubble separating two spacetime regions: an interior de Sitter region with vacuum energy density ρ_b , and an exterior Schwarzschild (or Schwarzschild-FRW) region with negligible energy density $\rho_0 \approx 0$ after inflation. The thin-wall formalism treats the bubble wall as a timelike hypersur-

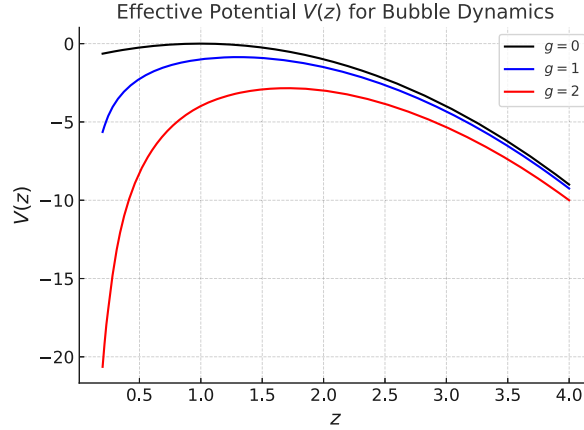


Fig. 1. Effective potential $V(z)$ governing the dynamics of a vacuum bubble for different values of the dimensionless parameter $g = 4H_\sigma/H_+$. The shape of the potential determines whether the bubble expands indefinitely (inflating interior) or recollapses (singularity).

face of surface tension σ , whose motion is governed by the balance between interior vacuum pressure, exterior geometry, and wall self-gravity. Explicitly, the junction condition reads

$$\sqrt{\dot{R}^2 + f_{\text{in}}(R)} - \sqrt{\dot{R}^2 + f_{\text{out}}(R)} = 4\pi G\sigma R,$$

where $f_{\text{in}}(R) = 1 - H_b^2 R^2$ is the de Sitter lapse function inside the bubble, $f_{\text{out}}(R) = 1 - 2GM/R$ is the Schwarzschild lapse function outside, and $R(\tau)$ is the bubble radius as a function of proper time τ . After squaring and rearranging, one obtains an effective energy conservation relation of the form of Eq. (1). Each term has a clear physical meaning: The first term, $\frac{4\pi}{3} R^3(\rho_b - \rho_0)$, represents the net vacuum energy difference between the inside and outside of the bubble. The second term, $4\pi\sigma R^2 \sqrt{\dot{R}^2 + 1 - H_b^2 R^2}$, corresponds to the contribution of the bubble wall itself, including its rest energy and kinetic energy. The final negative term, $-8\pi^2 G\sigma^2 R^3$, arises from the wall's self-gravity, which reduces the total effective mass as seen by an external observer. Thus Eq. (1) encapsulates the balance of vacuum pressure, wall tension, and gravitational backreaction. The fate of the bubble, whether collapse or eternal expansion, follows from analyzing this effective conservation equation.

Following the treatment in [18], the dynamics of the bubble wall can be expressed in the form:

$$\left(\frac{dz}{d\tilde{\tau}}\right)^2 + V(z) = E, \quad (2)$$

with the effective potential given by,

$$V(z) = -\left(1 - \frac{z^3}{z^2}\right)^2 - \frac{g^2}{z}, \quad (3)$$

where $g = 4H_\sigma/H_+$ and the dimensionless variables $z, \tilde{\tau}$ are defined by,

$$z^3 = \frac{H_+^2 R^3}{2GM}, \quad (4)$$

$$\tilde{\tau} = \frac{H_+^2 \tau}{4H_\sigma}, \quad (5)$$

with $H_+^2 = H_b^2 + 4H_\sigma^2$. The effective potential $V(z)$ shown in Fig. 1 governs the radial dynamics of a vacuum bubble wall in the thin-wall approximation.

To identify the transition between mentioned two behaviors (collapse, or unbounded expansion), we define a critical mass M_{cr} where the bubble just marginally inflates. This occurs when the potential $V(R)$ defined by the dynamics has a saddle point at the bubble's initial size. A reasonable estimate of M_{cr} is obtained by setting $\dot{R} = 0$ and solving,

$$M_{\text{cr}} = \frac{4\pi}{3} R^3(\rho_b - \rho_0) + 4\pi\sigma R^2 \sqrt{1 - H_b^2 R^2} - 8\pi^2 G\sigma^2 R^3, \quad (6)$$

requiring $R < H_b^{-1}$. The equality is saturated for bubbles on the verge of inflation. The inequality $M > M_{\text{cr}}$ thus defines the inflating regime.

By minimizing M_{cr} with respect to R , we find the critical radius R_c that optimizes the formation of a baby universe:

$$\left.\frac{dM_{\text{cr}}}{dR}\right|_{R=R_c} = 0 \quad \Rightarrow \quad R_c \approx \frac{1}{H_b}(1 - \epsilon), \quad \epsilon \ll 1. \quad (7)$$

This shows that baby universe formation becomes exponentially more probable for bubbles formed early during inflation, when $R_c \gg H_i^{-1}$.

In this work we use the term supercritical in a purely dynamical/causal sense tied to the thin-wall bubble dynamics. Concretely, let $M_{\text{cr}}(R; \rho_b, \rho_0, \sigma)$ denote the critical mass obtained from the junction-condition energy relation Eq. (1) and its stationary version Eqs. (6) and (7), which marks the onset of unbounded wall expansion and the formation of a wormhole to an inflating interior. Then

$$\text{supercritical} \iff M > M_{\text{cr}},$$

$$\text{subcritical} \iff M < M_{\text{cr}}$$

with M_{cr} fixed by the microphysical parameters $\{\rho_b, \rho_0, \sigma\}$ and the existence of an extremum in the effective potential for the wall. For $M < M_{\text{cr}}$ the wall turns around and collapses, yielding a Schwarzschild interior singularity; for $M > M_{\text{cr}}$ the solution evolves through a wormhole throat into an inflating (de Sitter-like) baby universe (Equivalently, the same threshold is seen by analyzing the effective potential $V(z)$ in Eqs. (2)–(5): supercritical solutions correspond to trajectories that avoid a turning point and escape to the inflating branch).

The designation supercritical does not imply exceeding some maximum mass or leaving the black-hole family. To external observers in the parent FRW region, the spacetime is that of a black hole with horizon radius $r_s = 2GM$ (possibly with small cosmological/FRW corrections). The term only characterizes the interior causal structure: for $M > M_{\text{cr}}$ there exists a (non-traversable) wormhole to an inflating region; for $M < M_{\text{cr}}$ the interior terminates at a singularity. In both cases the exterior is a bona fide black hole described by the mass parameter M (Operationally, M is the ADM/Schwarzschild mass measured outside the wall in Eq. (1); the wall tension and self-gravity shift this mass via the last two terms of Eq. (1)).

It should be noted that our usage is unrelated to supercritical phases in black-hole thermodynamics (e.g., the van der Waals-like critical point and supercritical region in extended P – V thermodynamics). We do not assume or invoke a thermal phase transition between small and large black holes. Instead, supercritical here refers to crossing the geometric/dynamical threshold M_{cr} that changes the global causal structure of the solution from collapse to internal inflation governed by the thin-wall junction conditions.

Black holes containing inflating interiors contribute less visible entropy than their true area would suggest. Let $A = 4\pi R_s^2$ be the Schwarzschild horizon area. The Bekenstein-Hawking entropy is given by,

$$S_{\text{BH}} = \frac{A}{4G} = \frac{\pi R_s^2}{G}. \quad (8)$$

However, when a baby universe forms, degrees of freedom become inaccessible to the parent universe. This leads to an entropy deficit:

$$\Delta S = S_{\text{BH}}^{\text{naive}} - S_{\text{eff}} = \gamma S_{\text{ent}}, \quad \gamma < 1, \quad (9)$$

where S_{ent} accounts for the entanglement entropy between the parent and baby universes. This modifies the effective entropy budget:

$$S_{\text{total}}^{\text{eff}} = S_{\text{BH}} + S_{\text{ent}}, \quad (10)$$

where $S_{\text{ent}} \sim \alpha \log \mathcal{N}$, with $\mathcal{N}$ denoting the number of accessible internal vacua in the baby universe landscape. This provides a microphysical justification for the correction term introduced in Section 3.

The ultimate fate of the bubble depends critically on its mass. If the initial mass parameter, effectively the energy deficit due to the displaced FRW matter combined with the wall tension and vacuum energy difference, exceeds a critical threshold $M > M_{\text{cr}}$, then the resulting black hole geometry possesses a wormhole structure. This wormhole connects the parent universe to a newly formed inflating region on the other side, resembling a baby universe that evolves internally like a separate de Sitter spacetime. The value of M_{cr} depends on the microphysical parameters of the wall, particularly its tension σ and the vacuum energies involved, and is typically on the order of the inverse of the larger of the two time scales, $t_b \sim H_b^{-1}$ or $t_\sigma \sim (2\pi G\sigma)^{-1}$ [18,40].

The internal inflation of the baby universe proceeds eternally if the vacuum ρ_b is metastable or absolutely stable. The geometry can be described by a black hole with a non-traversable wormhole, whose throat connects to a spacelike hypersurface on the inflating side. This scenario bears resemblance to previous models where black hole interiors were conjectured to harbor new universes [41–43]. However, the novelty in this mechanism lies in the natural cosmological production of such objects from quantum tunneling processes during inflation, as opposed to engineered or speculative initial conditions.

The baby universe, once nucleated, becomes causally disconnected from its parent spacetime. Nonetheless, quantum correlations established during the bubble's nucleation and collapse persist. These correlations are encoded in the entanglement structure of the initial state and motivate our later proposal that such black hole-induced baby universes contribute to the generalized entropy of the multiverse. The presence of a wormhole structure, even if classically non-traversable, supports the possibility of a quantum channel—akin to the ER=EPR conjecture [28]—whereby the parent and child regions remain entangled across the horizon.

Conversely, if the mass is subcritical ($M < M_{\text{cr}}$), the bubble fails to reach sufficient expansion and instead collapses entirely. The interior contracts, the bubble wall recedes, and the entire configuration evolves into a singularity within a Schwarzschild black hole. No baby universe is formed in this case, and the geometry is indistinguishable from that of a primordial black hole. The distinction between these outcomes hinges on the initial nucleation time, since bubbles nucleated earlier during inflation have more time to grow and are thus more likely to become supercritical [18,20].

An intriguing implication of this process is the production of a broad spectrum of black holes during inflation, some of which may harbor inflating interiors. This leads to a rich and complex global spacetime structure in which distinct inflating domains—baby universes—are generated within black holes scattered throughout the parent universe. These internal inflating regions can subsequently

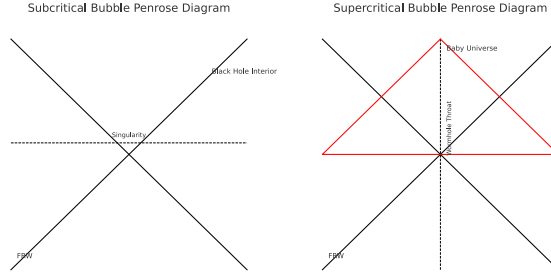


Fig. 2. Penrose diagrams illustrating the causal structure of spacetimes resulting from vacuum bubbles. **Left:** A subcritical bubble that collapses to form a Schwarzschild black hole with a central singularity. **Right:** A supercritical bubble leading to a black hole containing a baby universe connected via a non-traversable wormhole. The bubble wall lies near the wormhole throat, and the internal region undergoes eternal inflation.

undergo vacuum transitions of their own, spawning further generations of baby universes. The resulting multiverse is thus not a collection of disconnected bubbles on a single inflating background, but a hierarchically nested fractal of black holes and wormholes, each potentially containing further inflating regions [44,45].

From the standpoint of causal structure (see Fig. 2), each black hole of supercritical mass forms a Penrose diagram analogous to the extended Kruskal solution, with one exterior region replaced by an inflating cosmology. The matching of Schwarzschild, de Sitter, and FRW patches across domain walls has been analyzed in detail in previous works [21,38,46], and the wormhole throat typically closes at finite time from the perspective of external observers. After this pinch-off, the baby universe evolves autonomously, with no causal contact with the parent universe, though entanglement remains imprinted in the initial quantum state.

In what follows, we explore how such black hole-induced baby universes can be incorporated into a generalized thermodynamic framework, leading to a revised entropy accounting for the multiverse and a modified formulation of the second law that includes entanglement across disconnected regions.

3. Generalized entropy in the multiverse

We propose the following generalization of the Bekenstein-Hawking entropy:

$$S_{\text{total}} = S_{\text{BH}} + S_{\text{ent}}, \quad (11)$$

where $S_{\text{BH}} = \frac{kc^3 A}{4 G \hbar}$ is the usual entropy of the black hole, and S_{ent} is an entanglement entropy term accounting for the quantum correlations between the parent and daughter universes.

We posit that S_{ent} scales as:

$$S_{\text{ent}} \sim \alpha \log \mathcal{N}, \quad (12)$$

where $\mathcal{N}$ is the number of independent vacuum configurations accessible to the baby universe, and α is a dimensionless coupling parameter reflecting inter-universal decoherence strength.

We now provide a quantum information-theoretic derivation of the entropy flow across a black hole connecting a parent universe to a baby universe. Let the global quantum state of the system at the time of bubble formation be described by a pure state $|\Psi\rangle$ in a Hilbert space that factorizes as,

$$\mathcal{H} = \mathcal{H}_{\text{parent}} \otimes \mathcal{H}_{\text{baby}}. \quad (13)$$

Although the two subsystems become causally disconnected after the wormhole throat pinches off, their shared quantum origin ensures that they remain entangled. The reduced density matrix of the parent universe is obtained by tracing over inaccessible degrees of freedom in the baby universe:

$$\rho_{\text{parent}} = \text{Tr}_{\text{baby}} [|\Psi\rangle\langle\Psi|]. \quad (14)$$

The associated entanglement entropy is given by the von Neumann entropy:

$$S_{\text{ent}} = -\text{Tr} [\rho_{\text{parent}} \log \rho_{\text{parent}}], \quad (15)$$

which also equals the entropy of the baby universe due to purity of the global state: $S_{\text{ent}} = S_{\text{baby}}$.

As the bubble expands and the wormhole pinches off, the entanglement entropy grows, reaching a maximum when the Hilbert space dimensions of the two subsystems become comparable. This evolution mimics the well-known Page curve [47], and the entropy follows:

$$S_{\text{ent}}(t) \approx \begin{cases} \log \dim \mathcal{H}_{\text{baby}}(t) & \text{for } \dim \mathcal{H}_{\text{baby}} \ll \dim \mathcal{H}_{\text{parent}}, \\ \log \dim \mathcal{H}_{\text{parent}}(t) & \text{for } \dim \mathcal{H}_{\text{baby}} \gg \dim \mathcal{H}_{\text{parent}}. \end{cases} \quad (16)$$

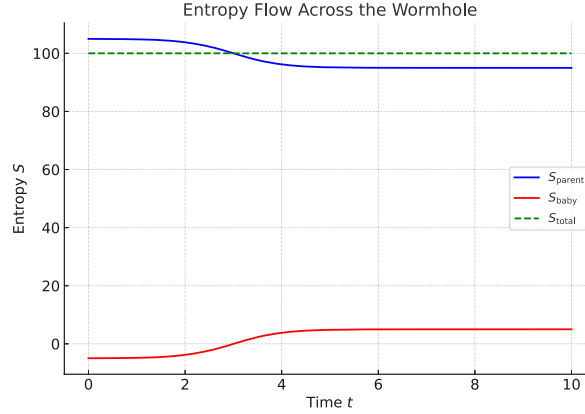


Fig. 3. Illustration of entropy flow across a black hole wormhole during baby universe formation. The entropy of the parent universe (S_{parent} , blue) decreases as internal degrees of freedom become hidden. The entropy of the baby universe (S_{baby} , red) increases due to entanglement. The total generalized entropy (S_{total} , green dashed) remains non-decreasing, consistent with a generalized second law. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

This implies that from the parent observer's perspective, entropy is not lost but transferred into hidden correlations with the baby universe. Consequently, the generalized entropy $S_{\text{BH}} + S_{\text{ent}}$ remains non-decreasing, preserving unitarity and satisfying a generalized second law. This framework thus resolves the apparent decrease in black hole entropy observed by external observers by accounting for quantum entanglement with baby universes.

To quantify the entanglement entropy S_{ent} , we estimate the effective Hilbert space dimension of the baby universe using the vacuum statistics of the string landscape. String theory compactifications, particularly in flux vacua scenarios such as those studied in [16,17,48–50], suggest that the number of metastable de Sitter vacua $\mathcal{N}$ is exponentially large, potentially as high as

$$\mathcal{N} \sim 10^{500}, \quad (17)$$

depending on the details of flux configurations, moduli stabilization, and topological data of the compactification manifold. If each vacuum defines a distinct semiclassical background with a corresponding quantum Hilbert space $\mathcal{H}_i$, the total Hilbert space dimension of the baby universe can be estimated as:

$$\dim \mathcal{H}_{\text{baby}} \sim \sum_{i=1}^{\mathcal{N}} \dim \mathcal{H}_i \sim \mathcal{N} \cdot e^{S_i}, \quad (18)$$

where $S_i \sim \pi/(GH_i^2)$ is the de Sitter entropy of vacuum i . Assuming typical de Sitter radii $H_i^{-1} \sim 10$ to 100 in Planck units, we find:

$$S_i \sim 10^2 - 10^4, \quad \Rightarrow \quad \log \dim \mathcal{H}_{\text{baby}} \gtrsim 10^5. \quad (19)$$

Hence, the entanglement entropy contributed by baby universes formed via black hole wormholes could be extremely large:

$$S_{\text{ent}} \sim \log \dim \mathcal{H}_{\text{baby}} \sim 10^5 \text{ or more.} \quad (20)$$

This entropy, though inaccessible to external observers, plays a vital role in conserving information globally. Its contribution dominates the apparent entropy loss due to black hole evaporation or internal inflation and supports a refined formulation of entropy bounds and thermodynamic consistency across the multiverse.

Fig. 3 illustrates the time evolution of entropy during the formation of a baby universe inside a supercritical black hole. In this process, quantum degrees of freedom that were initially accessible within the parent universe become sequestered behind the event horizon as the internal bubble undergoes inflation. The blue curve represents the entropy S_{parent} measured within the observable region of the parent universe. As the bubble wall recedes and the wormhole throat pinches off, a portion of the system's microstates becomes inaccessible to external observers, resulting in an effective decrease of observable entropy.

Conversely, the red curve shows the entropy S_{baby} associated with the internal inflating region. Although causally disconnected, the baby universe remains quantum mechanically entangled with its parent, and thus possesses an entropy contribution $S_{\text{ent}} \sim \alpha \log \mathcal{N}$, where $\mathcal{N}$ is the number of internal vacuum configurations. The green dashed curve depicts the total generalized entropy $S_{\text{total}} = S_{\text{parent}} + S_{\text{baby}}$, which remains non-decreasing, in accordance with a multiversal extension of the second law of thermodynamics. This behavior provides strong motivation for modifying traditional entropy bounds in spacetimes that include inflating interior regions connected by wormholes.

While the string landscape contains an enormous number of metastable vacua, not all of them are accessible to a given baby universe formed via black hole nucleation. Gravitational tunneling constraints—specifically, those arising from Coleman-DeLuccia (CDL) instanton solutions—limit the transitions between vacua based on the energy difference, tension of the bubble wall, and background curvature [51–53].

The CDL formalism implies that a vacuum ρ_b can only nucleate from a parent vacuum ρ_i if the tunneling action B is finite and the Euclidean bounce solution exists. In the thin-wall approximation, the action takes the form:

$$B = \frac{27\pi^2\sigma^4}{2(\Delta\rho)^3}, \quad (21)$$

where $\Delta\rho = \rho_i - \rho_b$ and σ is the wall tension. For transitions from a high-energy vacuum to a much lower one, the action becomes exponentially large, suppressing the tunneling rate:

$$\Gamma \sim e^{-B}. \quad (22)$$

Moreover, transitions into vacua with $\rho_b < 0$ (AdS) typically lead to big crunch singularities, and are not expected to produce inflating baby universes [54]. Therefore, only vacua with small positive ρ_b and favorable tension-to-energy ratios contribute meaningfully to the Hilbert space accessible to a given bubble. Let $\mathcal{N}_{\text{eff}} \subset \mathcal{N}$ be the subset of vacua satisfying CDL accessibility conditions:

$$\mathcal{N}_{\text{eff}} \sim \mathcal{N} \cdot f_{\text{CDL}}, \quad f_{\text{CDL}} \ll 1. \quad (23)$$

Estimates from landscape studies suggest $f_{\text{CDL}} \sim 10^{-3}$ to 10^{-100} , depending on moduli stabilization and flux compactification details [45,50]. As a result, the effective Hilbert space dimension becomes:

$$\dim \mathcal{H}_{\text{baby}}^{(\text{eff})} \sim \mathcal{N}_{\text{eff}} \cdot e^{S_i}, \quad \Rightarrow \quad S_{\text{ent}}^{(\text{eff})} \sim \log \mathcal{N}_{\text{eff}} + \langle S_i \rangle. \quad (24)$$

Thus, while the raw number of string vacua implies a massive entanglement entropy, physical constraints reduce the effective contribution. Nonetheless, even for conservative values of $\mathcal{N}_{\text{eff}} \sim 10^{100}$, one finds:

$$S_{\text{ent}}^{(\text{eff})} \gtrsim 10^2 + 100 \log 10 \sim 330, \quad (25)$$

which is still significant when compared to the Bekenstein-Hawking entropy of primordial or inflationary black holes. This refinement underscores that the entanglement entropy attributed to baby universes is not only large but physically regulated by vacuum dynamics. These constraints are essential in computing entropy flow in a consistent multiverse framework.

4. The multiversal second law

The classical second law of thermodynamics asserts that the entropy of a closed system must not decrease with time. In standard cosmology, this law applies within the bounds of a causal patch, typically bounded by an event horizon. However, in a multiversal framework where black hole-induced inflation gives rise to causally disconnected baby universes, the traditional formulation is no longer sufficient. In this context, the notion of entropy must be broadened to include both local and nonlocal contributions arising from quantum entanglement across different spacetime domains.

We thus propose a multiversal extension of the second law, expressed as:

$$\frac{d}{dt} \left(S_{\text{total}}^{\text{univ}} + \sum_i S_{\text{ent}}^{(i)} \right) \geq 0, \quad (26)$$

where $S_{\text{total}}^{\text{univ}}$ is the coarse-grained entropy accessible to observers within the parent universe, and the sum over $S_{\text{ent}}^{(i)}$ captures the entanglement entropy associated with baby universes i formed via black hole nucleation and internal inflation.

This formulation acknowledges that the universe, when embedded in a broader multiversal structure, is not thermodynamically closed in the traditional sense. Instead, it undergoes entropy exchange with its hidden counterparts through the formation of wormhole-connected regions. Although these baby universes are causally disconnected after the pinch-off of the wormhole throat, quantum correlations persist due to their shared origin in a pure state. These correlations encode information about the internal microstates of the baby universe and must be counted in the global entropy budget to maintain unitarity.

The multiversal second law thus ensures that the total generalized entropy—including both observable and hidden contributions—never decreases. This reconciles the apparent paradox of entropy loss in black hole evaporation and internal inflation. From the viewpoint of an external observer, the entropy of a black hole may appear to decrease as internal degrees of freedom inflate beyond observational reach. However, the entanglement entropy with the hidden inflating region compensates for this loss, maintaining a globally non-decreasing entropy trajectory.

Moreover, the growth of entanglement entropy is tied to the dynamical expansion of the Hilbert space of the baby universe. As it inflates and explores a broader range of internal vacuum configurations, the effective number of microstates increases, leading to a monotonic rise in $S_{\text{ent}}^{(i)}$. This growth follows a Page-curve-like evolution and saturates once the Hilbert space of the baby universe becomes comparable in dimensionality to that of the parent region. The result is a thermodynamic arrow of time that is consistent across multiversal branches.

This generalized law not only preserves the spirit of the second law in the presence of horizon-crossing degrees of freedom but also suggests a fundamental principle of quantum gravity: that entropy production is a global property of spacetime and cannot be confined to individual causal patches when entanglement connects them. It supports a nonlocal perspective in which spacetime regions share information through wormhole geometries, aligning with the ER=EPR paradigm.

Importantly, this framework implies that black hole formation and internal inflation are not entropic dead ends, but rather, active sites of entropy redistribution in the multiverse. Each supercritical black hole acts as a thermodynamic bridge to a new inflating

region, contributing positively to the overall entropy balance. The multiversal second law therefore provides a robust and consistent arrow of time for a universe embedded in a branching, entangled cosmic structure.

To quantify the generalized second law in a dynamical context, we introduce a time-dependent form of the total entropy in the presence of $N(t)$ baby universes formed up to time t :

$$S_{\text{gen}}(t) = S_{\text{univ}}(t) + \sum_{i=1}^{N(t)} S_{\text{ent}}^{(i)}(t), \quad (27)$$

where $S_{\text{univ}}(t)$ is the observable coarse-grained entropy of the parent universe, and $S_{\text{ent}}^{(i)}(t)$ denotes the entanglement entropy of the i -th baby universe.

Assuming each baby universe inflates eternally and its Hilbert space grows with time, we model its entropy as:

$$S_{\text{ent}}^{(i)}(t) \approx \beta \log \dim \mathcal{H}_i(t), \quad (28)$$

where $\mathcal{H}_i(t)$ denotes the Hilbert space of the i -th inflating domain, and β encodes the entanglement strength per accessible state. For a landscape of de Sitter vacua, $\dim \mathcal{H}_i(t)$ can grow exponentially with the number of accessible vacua and Hubble volume.

Assuming a typical growth rate $\dot{S}_{\text{ent}}^{(i)} \propto H_i^3$ due to vacuum expansion, and a formation rate $\Gamma(t)$ of new baby universes per unit time (per comoving volume), we can write,

$$\frac{d}{dt} S_{\text{gen}}(t) = \frac{d S_{\text{univ}}}{dt} + \int_0^t \Gamma(\tau) \frac{d S_{\text{ent}}^{(\text{new})}(\tau)}{d\tau} d\tau + \Gamma(t) S_{\text{ent}}^{(\text{new})}(t), \quad (29)$$

where the first integral accounts for the cumulative entropy growth in existing baby universes, and the last term includes the instantaneous entropy jump from newly formed inflating domains.

If we assume each new baby universe contributes a roughly constant entanglement entropy $S_{\text{ent}}^{(\text{new})} \approx \bar{S}_{\text{ent}}$, and that $\Gamma(t)$ is slowly varying (e.g., during post-inflationary epochs), then:

$$\frac{d}{dt} S_{\text{gen}}(t) \gtrsim \Gamma(t) \bar{S}_{\text{ent}}, \quad (30)$$

implying a lower bound on entropy production sourced by the birth of entangled inflating regions.

Let us define the Hilbert space dimension of the parent universe as $\dim \mathcal{H}_{\text{univ}}$, and that of all baby universes combined as:

$$\dim \mathcal{H}_{\text{baby}}(t) = \prod_{i=1}^{N(t)} \dim \mathcal{H}_i(t). \quad (31)$$

The entanglement entropy between the two regions is then bounded by the minimum dimension:

$$S_{\text{ent}}(t) \leq \log \min(\dim \mathcal{H}_{\text{univ}}, \dim \mathcal{H}_{\text{baby}}(t)). \quad (32)$$

This inequality yields a multiversal Page bound:

$$S_{\text{gen}}(t) \leq S_{\text{univ}}^{\text{max}} + \log \dim \mathcal{H}_{\text{univ}}, \quad (33)$$

which reflects the saturation of entanglement entropy once the baby universes together explore a Hilbert space larger than that of the parent universe.

In this regime, further entropy increase from entanglement halts, and only classical thermodynamic processes in the parent region can increase S_{gen} . This bound ensures consistency with unitarity and entropy conservation across multiversal branches.

Fig. 4 illustrates the time evolution of entropy components in a multiverse scenario. The blue curve represents the entropy $S_{\text{univ}}(t)$ of the parent universe, which increases slowly due to local thermodynamic processes, modeled here as logarithmic growth. The red

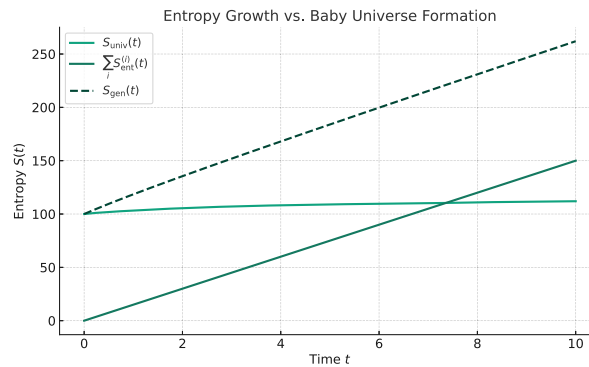


Fig. 4. Time evolution of entropy components in a multiverse scenario. The parent universe entropy $S_{\text{univ}}(t)$ grows slowly via standard thermodynamic processes, while the entanglement entropy $\sum_i S_{\text{ent}}^{(i)}(t)$ increases linearly with the rate of baby universe formation. The total generalized entropy $S_{\text{gen}}(t)$ remains non-decreasing, consistent with the multiversal second law.

curve shows the cumulative entanglement entropy $\sum_i S_{\text{ent}}^{(i)}(t)$ arising from the formation of baby universes through black hole-induced inflation. This contribution grows approximately linearly with time, assuming a constant formation rate and typical entanglement entropy per universe. The dashed green curve shows the total generalized entropy $S_{\text{gen}}(t) = S_{\text{univ}}(t) + \sum_i S_{\text{ent}}^{(i)}(t)$, which remains manifestly non-decreasing. This behavior supports the multiversal extension of the second law, showing that even if observable entropy in a causal patch stagnates or appears to decline (e.g., during black hole evaporation), the hidden entanglement with inflating regions ensures the global entropy continues to rise. The linear dominance of $S_{\text{ent}}^{(i)}$ at late times further emphasizes that baby universe formation becomes the primary driver of entropy production in an eternally branching multiverse.

We can further formalize the generalized second law using a continuity equation for entropy flow in the multiverse. Let $\rho_S(x, t)$ denote the entropy density in a parent universe region x , and $j_S^i(x, t)$ the entropic flux into baby universe i . Then:

$$\frac{\partial \rho_S(x, t)}{\partial t} + \nabla \cdot \vec{j}_S(x, t) = - \sum_i \delta(x - x_i) \Gamma_i(t) S_{\text{ent}}^{(i)}(t), \quad (34)$$

where $\Gamma_i(t)$ is the rate of baby universe formation at location x_i . This formalism captures the local entropic sink due to horizon-exceeding degrees of freedom and embeds entanglement into classical thermodynamic flow equations.

5. Implications and observational prospects

The framework developed in this work offers not only a theoretical unification of entropy dynamics across disconnected cosmological domains, but also a pathway toward observational probes of the multiverse through black hole physics. In particular, the existence of black holes whose masses cluster near the critical threshold M_{cr} may hint at internal inflating geometries. These supercritical black holes could serve as seeds of baby universes, and their abundance and properties may leave imprints in astrophysical and cosmological data.

5.1. Gravitational wave echoes and wormhole geometries

A promising avenue involves the analysis of post-merger gravitational wave signals from black hole collisions. Several studies have proposed that exotic compact objects—such as those containing internal structure or modified horizon physics—may exhibit gravitational wave echoes following the main ringdown phase. If the remnant black hole hosts a wormhole leading to an inflating baby universe, then partial reflections of perturbations off the wormhole throat may give rise to delayed secondary signals.

These echoes would be quasi-periodic and damped, with characteristic timescales determined by the wormhole geometry and throat location. The absence of classical reflectivity at the horizon in standard general relativity makes any such echo signature a compelling signal of quantum gravitational structure. Advanced LIGO, Virgo, and upcoming detectors such as LISA or Einstein Telescope could place stringent bounds or potentially detect such echoes.

5.2. Primordial black hole populations and critical mass peaks

Another observational window comes from the mass distribution of primordial black holes (PBHs). During inflation, quantum fluctuations in the energy density can lead to the formation of vacuum bubbles, some of which collapse into black holes. The model predicts a critical mass threshold M_{cr} separating those that collapse entirely from those that inflate internally. A statistical excess or sharp feature in the PBH mass spectrum near M_{cr} could indicate a phase transition between these two regimes.

Surveys across gravitational lensing, microlensing, and accretion signatures (e.g., via Subaru-HSC, EROS, or upcoming Roman Space Telescope) can constrain the PBH population. Any observed clustering of black holes near a particular mass scale, especially in regions of low baryonic feedback, may reflect the multiversal dynamics described in this paper.

5.3. Entropy bounds, holography, and quantum non-locality

From a more fundamental perspective, the conservation of generalized entropy across universes implies a form of quantum non-locality—a global structure that persists beyond causal disconnection. If baby universes remain entangled with their progenitor regions, then violations of traditional entropy bounds (such as the covariant Bousso bound) might occur locally, while still respecting the multiversal second law globally.

In principle, this could manifest in cosmological observables. Deviations from expected holographic scaling of entropy in cosmological voids or horizons might be observable via CMB constraints on late-time entropy production, thermalization of relics, or mismatch between energy flux and entropy change. Moreover, tensor modes from inflation could carry subtle imprints of the bubble nucleation network, especially if wormhole geometries briefly affect horizon-crossing modes.

5.4. Quantum correlations across universes

If the ER=EPR framework is correct, then quantum entanglement between parent and baby universes could have implications for the structure of quantum field correlations near black hole horizons. While direct detection of such correlations is likely beyond current technology, holographic dualities—such as AdS/CFT analogs or tensor network models—may allow simulation of entangled multiverse branches in controlled setups.

Table 1

Summary of observational channels relevant to multiversal entropy dynamics and baby universe formation.

Observational Channel	Potential Signature
Gravitational Wave Echoes	Delayed post-merger signals from black holes with wormhole interiors
Primordial Black Hole Mass Function	Mass distribution peaking near M_{cr} indicating onset of internal inflation
Cosmic Microwave Background (CMB)	Deviations from standard entropy bounds or thermal relic anomalies
Large-Scale Structure and Voids	Unusual entropy scaling or energy-entropy mismatch in underdense regions
Holographic Duals / Simulations	Indirect signatures of inter-universal entanglement via tensor network or AdS/CFT analogs
Vacuum Energy Anomalies	Apparent late-time dark energy shifts due to entropy transfer to hidden sectors

Furthermore, recent proposals for probing entanglement wedge reconstruction in cosmological or FRW spacetimes may open a route to inferring entanglement structure between disconnected domains from observable data on one side. If the entanglement entropy S_{ent} grows sufficiently fast, it may backreact on the effective dynamics of the parent universe, producing testable anomalies in late-time cosmological expansion or vacuum energy.

5.5. Experimental and theoretical frontiers

Ultimately, the observational prospects of this framework rely on improved detection sensitivity across gravitational wave channels, black hole population surveys, and deep cosmic probes. Future missions such as LISA, the Square Kilometre Array (SKA), and CMB-S4 may offer the precision needed to distinguish standard cosmological evolution from scenarios involving entropy transfer to baby universes.

On the theoretical side, simulations of black hole interiors with internal inflation, quantum circuit models of multiversal entanglement, and holographic RG flows incorporating bubble nucleation dynamics may offer deeper insight. The multiversal second law could become a cornerstone of a generalized thermodynamics of quantum spacetime, unifying locality, information, and inflationary structure under a single entropic principle.

A summary of potential observational channels and their expected signatures is given in [Table 1](#).

6. Conclusion

We have proposed a framework in which supercritical black holes serve as conduits for the creation of new inflating domains—baby universes—embedded within a multiversal structure. These regions, though causally disconnected after wormhole pinch-off, remain quantum mechanically entangled with their progenitor universe. This entanglement modifies the traditional notion of entropy, requiring the inclusion of hidden correlations in the total entropy budget.

By incorporating the entanglement entropy between parent and child universes, we derived a generalized second law of thermodynamics that remains valid even in the presence of horizon-crossing degrees of freedom. This *multiversal second law* ensures that global entropy increases due to both conventional thermodynamic processes and the formation of entangled inflating domains beyond causal contact.

Our analysis extends the Bekenstein-Hawking entropy to include entropic contributions from quantum entanglement across space-time regions, and places the dynamics of black hole interiors in a broader cosmological context. Black hole-induced inflation, in this view, is not an endpoint but a branching mechanism through which the multiverse grows and stores information.

We also outlined potential observational signatures of this framework, including gravitational wave echoes, critical mass distributions of primordial black holes, and violations of local entropy bounds—each of which may serve as indirect probes of the underlying multiversal structure.

This perspective unifies gravitational thermodynamics, quantum information, and cosmological inflation under a single principle of generalized entropy conservation. It offers a novel resolution to the black hole information paradox and opens a pathway toward testable predictions linking entanglement, geometry, and the origin of the multiverse.

Future work may focus on more precise modeling of baby universe Hilbert spaces, the dynamics of wormhole geometries in semi-classical gravity, and embedding this scenario within a holographic or quantum gravitational framework. The multiversal second law may ultimately be a foundational principle in the thermodynamics of quantum spacetime.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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